is that the National Milk Testing Service with its 350 staff should be disbanded and its work handed over to the Milk Marketing Board and the Veterinary Inspectorate. It was always difficult to see any real justification for the existence of the National Milk Testing Service.

In conclusion, one feature of the reorganization planned by the Wilson Committee may be pointed out which will please those who, like the writer, continue to maintain that the advisory and educational services should never have been divorced and should be 'remarried' as soon as possible. The County Agricultural Committees will remain based, in general, on existing counties and will have, as their chief concern, technical development and advisory work. It would be a small and comparatively easy step to add to these duties the responsibility for agricultural education in the counties. A great accession of strength would accrue to both services and further large economies in man-power and expenditure could be achieved together with greatly increased efficiency.

H. W. Gardner

STRUCTURE OF VITAMIN B_{12}

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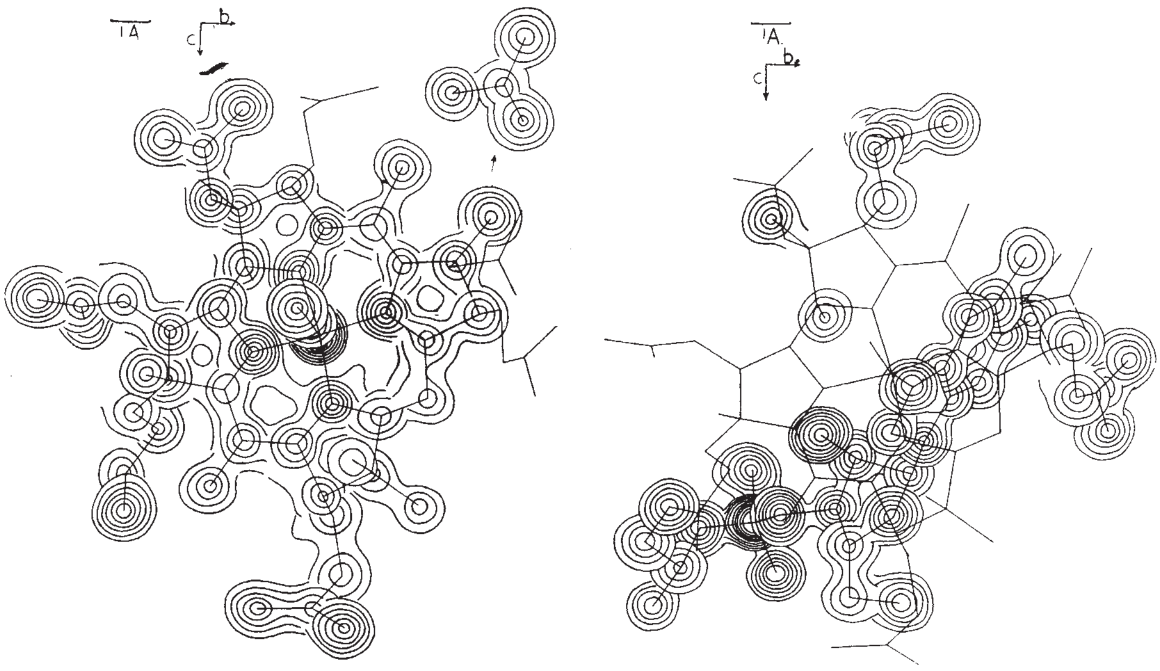
Since our communication in August 1955[1], we have carried our refinement of the structure of vitamin B_{12} a critical stage further. Four more calculations of the electron density distribution have been made, one for the wet B_{12} crystals, two for air-dried B_{12} crystals and one for the hexacarboxylic acid.

In Fig. 1 are shown the electron density peaks over all the atoms of the B_{12} molecule as it appears from our present calculations in the wet crystals. These peaks are rather lower than those observed for the hexacarboxylic acid owing to the smaller number of X-ray reflexions given by the larger molecule in proportion to its size. But they serve to show the relative positions of all the atoms in space of this very complex molecule within probably as little as 0·3 A. Similar evidence has been obtained for air-dried B_{12}; for both crystals the reliability indices in the latest structure factor calculations, 36·1 and 30·6 per cent, respectively, are now low



(a) (b)

Fig. 1. Electron density-levels from the three-dimensional electron density distribution for wet B_{12} crystals, calculated with terms phased on 93 atoms

Here, and in Fig. 4, the contours are drawn in the sections of the calculated distribution parallel with the *a* plane, passing at or near the atomic positions. Owing to the complexity of the molecule, complete contours for every atom are not given and the figure is divided into two parts. (*a*) shows the cobalt-containing nucleus and cyanide group; (*b*) the benziminazole, sugar, phosphate and propanolamine groups, the side chains being divided between the two. The acetamide chain on ring *B* is inset. The contour interval is 1 $e/A.^3$

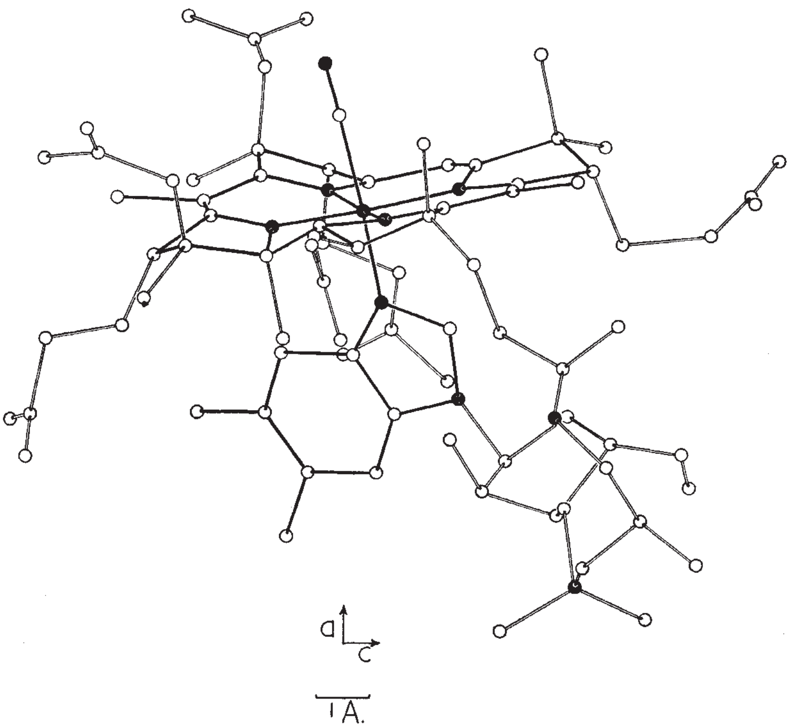
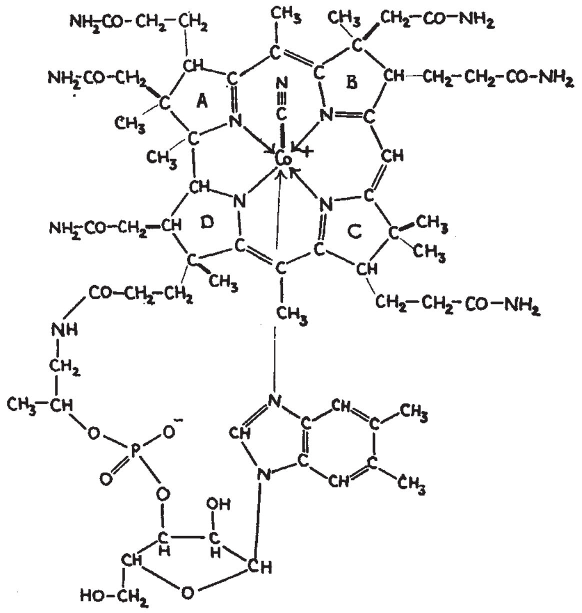


Fig. 2. The atomic positions found in the molecule of vitamin B_{12}. These are shown as derived in the wet crystals, projected on the *b* plane. To distinguish different parts of the molecule, bonds within the cobalt containing nucleus, benziminazole and cyanide groups are shown in black, together with the cobalt and phosphorus atoms and the nitrogen atoms of the nucleus, benziminazole, propanolamine and cyanide groups

enough to indicate that the solution we have reached is likely to be essentially correct.

The molecule that appears is very beautifully composed, not far from spherical in form, with all the more chemically reactive groups on its surface. It is drawn in projection in Fig. 2. It is built around the two planes of the benziminazole nucleus and the central cobalt-containing nucleus, which are nearly at right angles to each other. The ribose ring turns in a position nearly normal to the benziminazole group, which permits its easy linking through the phosphate, propanolamine, and propionic acid residues to ring *D* of the planar group. The benziminazole nucleus is packed in on either side by the propionamide side-chains in the extended, staggered configuration, attached to rings *A* and *B*; the side-chain on ring *C*, in the gauche configuration, lies above the ribose ring. All the acetamide residues project from the opposite side of the nucleus to the propionamide residues, towards the cyanide group. It is an interesting point that one of them—that on ring *B*—swings round the carbon–carbon single bond from a position directed away from, to one in contact with, the cyanide group, when the crystals are removed from their mother liquor; this permits the rather closer packing of the molecules found in the air-dried crystals. Throughout the molecule, as shown by Figs. 1 and 2, the atomic positions found conform in a most convincing way with the stereochemical rules established by the study of simpler molecules.

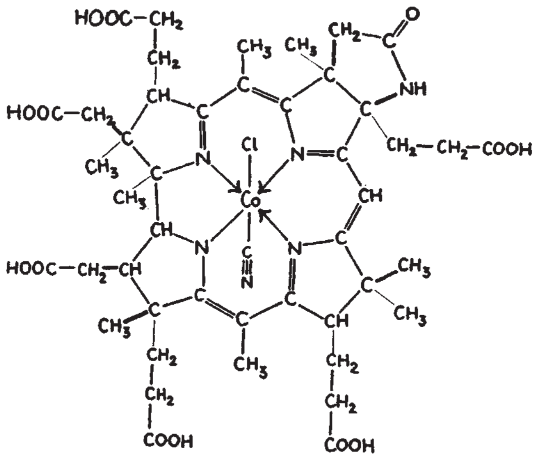
It now seems reasonably certain that we should formulate vitamin B_{12}, $C_{63}H_{88}O_{14}N_{14}PCo$, as in I, and the hexacarboxylic acid as in II, with six double bonds in the inner ring of the central nucleus; these can form a resonating system by intervention of the cobalt atom as illustrated by III. Some evidence in support of this has been obtained by theoretical calculations[2], by spectroscopic measurements[3] and from further chemical studies[4]. Our own evidence for this formulation is the geometrical form of the nucleus, which is illustrated by Fig. 3, and which is closely similar in all three crystals studied. The interatomic distances shown in this figure are derived from the latest calculations on the hexacarboxylic acid and are not individually very precise; in other parts of the molecule distances differing by as much as 0·16 A. from accepted bond-lengths have been found. But it is notable that in the region N_6–N_{18} *all* the bond-lengths are shorter than normal single bond distances; their average value, 1·36 A., is similar to that found in the phthalocyanines[5]. The positions of C_{29} and C_{30}, one or other of which would be expected to be out of the plane of the central ring system if only five double bonds were present, are even more significant. In the hexacarboxylic acid they appeared at all stages of the refinement to be within 0·2 A. of this plane, but in the B_{12} crystals themselves their positions were at first more confused, particularly C_{29}. As a critical test, accordingly, this atom was omitted from the phasing calculations used for the latest electron distribution calculated for air-dried B_{12}. It has now appeared, as Fig. 4 shows, as a small but clearly defined peak at a site which is closely equivalent to that found in the hexacarboxylic



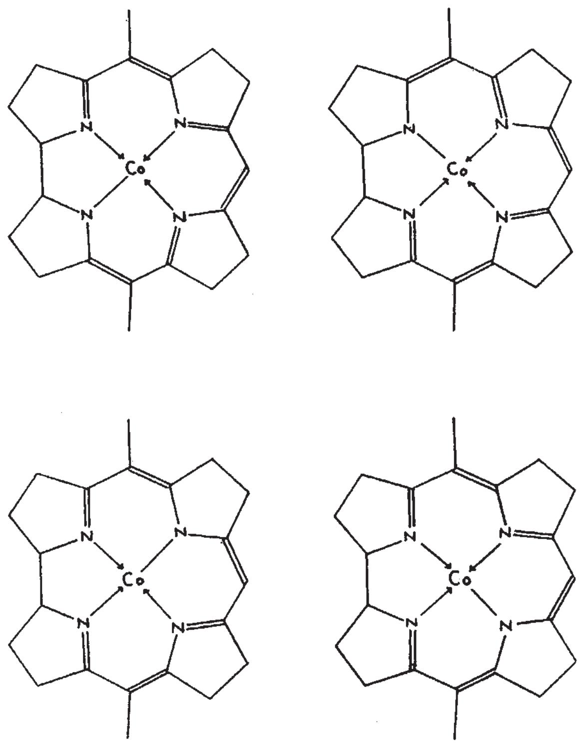
(I)

acid along the line Co to C_8. There is still some departure from planar character in this region of the inner nucleus in both wet and air-dried crystals, but this seems to affect N_{10}, C_9 and C_8 as much as C_{29}. If real, it may be a consequence of a small amount of compression from the benziminazole nucleus, which is in contact with these atoms. It is, in any event, not of the geometrical form that would be expected if C_8 were reduced.

The crystal structures of the different B_{12} compounds are still not completely solved; many of the water molecules which occupy space between the vitamin molecules have been placed, but not all. Still further small adjustments are to be expected on continued refinement of the atomic positions observed. We hope that future calculations will add still greater precision and certainty to our knowledge of the structure of the molecule.



(II)



(III)

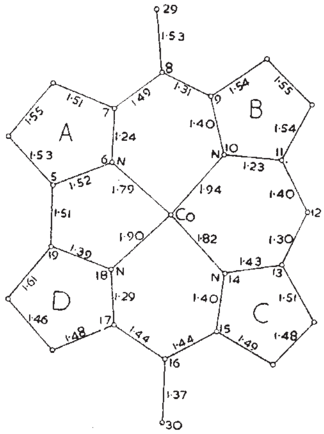


Fig. 3. Interatomic distances found in the inner nucleus of the hexacarboxylic acid, derived from nine refinements of the calculated electron density distribution

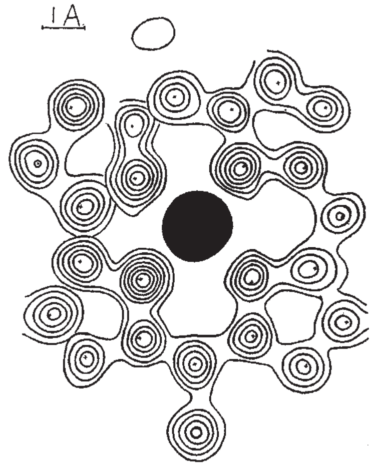


Fig. 4. Electron density peaks over the nucleus in air-dried vitamin B_{12}, derived from the three-dimensional electron density distribution calculated with terms phased on 95 atoms (Princeton data)

The latest series of calculations on wet B_{12} were carried out on the University of Manchester electronic computer through the generous co-operation of Dr. D. W. J. Cruickshank, Miss Diana Pilling and Mr. J. F. P. Donavan, of the University of Leeds. Those on air-dried B_{12} and on the hexacarboxylic acid were made on the National Bureau of Standards automatic computer in Los Angeles, supported by the United States Office of Naval Research.

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